

1. Basics of Solidity

- Introduction to Solidity
 - Structure of a Solidity Program (`pragma`, `contract`, etc.)
 - Data Types (uint, int, bool, address, string, bytes)
 - Variables (state, local, global)
 - Comments and Naming Conventions
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2. Functions

- Function declaration and syntax
 - Function visibility: `public`, `private`, `internal`, `external`
 - Function modifiers: `view`, `pure`, `payable`
 - Return values and multiple returns
 - Function overloading
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3. Control Structures

- `if`, `else if`, `else`
 - `for` loop, `while` loop, `do-while` loop
 - `break` and `continue`
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4. Arrays

- Fixed-size arrays
 - Dynamic arrays
 - Push, pop, and delete operations
 - Looping through arrays
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5. Structs

- Defining and using structs
 - Storing structs in arrays
 - Structs inside mappings
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6. Mappings

- Syntax and usage
 - Nested mappings
 - Mapping with structs
 - Mapping vs Arrays
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7. Sending and Receiving Ether

- `payable` keyword
 - `msg.sender` and `msg.value`
 - `transfer`, `send`, and `call` methods
 - `receive()` and `fallback()` functions
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8. Error Handling

- `require()`
 - `assert()`
 - `revert()`
 - Custom Errors
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9. Events and Logging

- Declaring and emitting events
 - Using events to track transactions on blockchain
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10. Modifiers

- Custom modifiers
 - `onlyOwner` pattern
 - Using modifiers for access control
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11. Inheritance

- Single and multiple inheritance
 - `super` keyword
 - Function overriding
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12. Abstract Contracts & Interfaces

- Abstract contracts
 - Interfaces
 - Difference between both
 - Using interfaces to interact with other contracts
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13. Enums

- Defining enums
 - Using enums inside structs or functions
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14. Constructor & Ownership

- `constructor` function
 - Setting and checking owner
 - Ownership transfer
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15. Access Control

- Role-based access control
 - `Ownable` concept
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16. Libraries

- Internal vs External libraries
 - Using `using for` syntax
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17. Storage, Memory, and Calldata

- Difference between them
 - How to use in functions
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18. Gas Optimization

- Gas cost of operations
 - Using `constant` and `immutable`
 - Efficient data structures
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19. Blockchain Variables

- `msg.sender`, `msg.value`, `block.timestamp`, `block.number`, etc.
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20. Advanced Topics

- Delegatecall and Call
 - Low-level interactions
 - Contract-to-contract calls
 - Fallback mechanisms
 - Reentrancy and Security
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21. Projects / Practical Concepts

- Counter Contract
- Storage Contract
- Ether Wallet
- Voting System
- Crowd Funding
- NFT Minting
- ERC20 Token Contract